AI Champions Phase 1 - Appendix Q9

System architecture for UMBRAMED's hybrid clinical LLM and deterministic safety layer.

Data inputs	Hybrid reasoning core	Decision outputs
Diraya retrospective data - patient notes - lab results - medication history - protocol labels	1. Clinical LLM Mistral/Llama fine-tuning 2. Patient state encoder Context extraction 3. Deterministic safety engine Interactions, contraindications, dosing and guideline checks	Physician-facing output only if: - renewal is clinically appropriate - zero safety-critical violations - uncertain cases escalated to review

Validation benchmark matrix

Dimension	Baseline	Target Phase 1 result	Evidence source
Renewal appropriateness accuracy	Prototype 89%	>95% on complex cases	Retrospective 6,500-patient validation set
Safety-critical false negatives	Unknown	Zero	Deterministic engine plus held-out contraindication suite
Interaction detection sensitivity	Rule-based only systems vary	100% on known contraindications and interactions	DrugBank, AEMPS, BNF cross-check
Operational relevance	Alerts only	Actionable renewal recommendation with physician oversight	Workflow evaluation against Diraya renewal process

Design principle: the LLM never ships a recommendation directly to the clinician. Every output must independently pass the deterministic verification layer. If the rule engine is uncertain or finds a conflict, the case is forced into physician review.